ISEN 637 Lecture 10

Actor-Critic Algorithms, Policy Gradient (Infinite Horizon)

Outline

1. Actor-Critic Algorithm
2. Advantage Actor-Critic
3. Policy Gradient for Infinite Horizon Case

Actor-Critic Algorithms: A Closed-loop Policy Gradient

For a randomized (parametrized) policy π_θ, the value function

$$V_\theta^T(s) = \mathbb{E}_{a_t \sim \pi_\theta(\cdot|s_t), s_{t+1} \sim P(\cdot|s_t,a_t)}[\sum_{t=0}^{T} \gamma^t R(s_t, a_t) | s_0 = s]$$

$$:= \mathbb{E}_\theta[\sum_{t=0}^{T} \gamma^t R(s_t, a_t) | s_0 = s]$$

can be written as

$$V_\theta^T(s) = \sum_{a \in \mathcal{A}} \pi_\theta(a|s) Q_\theta^T(s,a) \tag{1}$$

where

$$Q_\theta^T(s,a) = R(s,a) + \gamma \sum_{s' \in \mathcal{S}} P(s'|s,a) V_\theta^{T-1}(s') \tag{2}$$

A Closed-loop Policy Gradient

Taking derivatives on (7) when evaluated in $s = s_0$ we obtain:

$$\begin{aligned}\nabla V_\theta^T(s_0) &= \sum_{a\in\mathcal{A}} [\nabla \pi_\theta(a|s_0) Q_\theta^T(s_0,a) + \pi_\theta(a|s_0)\nabla Q_\theta^T(s_0,a)] \\ &= \sum_{a\in\mathcal{A}} [\nabla \log \pi_\theta(a|s_0) Q_\theta^T(s_0,a)\pi_\theta(a|s_0) + \pi_\theta(a|s_0)\nabla Q_\theta^T(s_0,a)] \\ &= \mathbb{E}_\theta[\nabla \log \pi_\theta(a_0|s_0) Q_\theta^T(s_0,a_0)] + \sum_{a\in\mathcal{A}} \pi_\theta(a|s_0)\nabla Q_\theta^T(s_0,a)\end{aligned} \tag{3}$$

and since $\nabla_\theta R(s,a) = 0$ from (2)

$$\nabla Q_\theta^T(s_0,a) = \gamma \sum_{s_1\in\mathcal{S}} P(s_1|s_0,a)\nabla V_\theta^{T-1}(s_1) \tag{4}$$

A Closed-loop Policy Gradient

Hence substituting (4) in (3) we obtain:

$$\begin{aligned}\nabla V_\theta^T(s_0) &= \mathbb{E}_\theta[\nabla \log \pi_\theta(a_0|s_0) Q_\theta^T(s_0,a_0)] + \gamma \sum_{s_1 \in \mathcal{S}} \sum_{a \in \mathcal{A}} \pi_\theta(a|s_0) P(s_1|s_0,a) \nabla V_\theta^{T-1}(\\ &= \mathbb{E}_\theta\left[\nabla \log \pi_\theta(a_0|s_0) Q_\theta^T(s_0,a_0) + \gamma \nabla V_\theta^{T-1}(s_1)\right] \qquad (5)\end{aligned}$$

A Closed-loop Policy Gradient

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In a similar manner to (3) we can show:

$$\nabla V_\theta^{T-1}(s_1) = \mathbb{E}_\theta[\nabla \log \pi_\theta(a_1|s_1) Q_\theta^{T-1}(s_1,a_1)] + \sum_{a \in \mathcal{A}} \pi_\theta(a|s_1) \nabla Q_\theta^{T-1}(s_1,a) \tag{6}$$

A Closed-loop Policy Gradient

Substituting (6) into (5) we obtain:

$$\nabla V_\theta^T(s_0) = \mathbb{E}_\theta\big[\nabla \log \pi_\theta(a_0|s_0) Q_\theta^T(s_0, a_0) + \gamma \nabla \log \pi_\theta(a_1|s_1) Q_\theta^{T-1}(s_1, a_1) + \gamma^2 \nabla V_\theta^{T-2}(s_2)\big]$$

A Closed-loop Policy Gradient

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$$\nabla V_\theta^T(s_0) = \mathbb{E}_\theta\big[\nabla \log \pi_\theta(a_0|s_0) Q_\theta^T(s_0, a_0) + \gamma \nabla \log \pi_\theta(a_1|s_1) Q_\theta^{T-1}(s_1, a_1) + \gamma^2 \nabla V_\theta^{T-2}(s_2)\big]$$

Following this argument we obtain:

$$\nabla V_\theta^T(s_0) = \mathbb{E}_\theta\Big[\sum_{t=0}^{T-1} \gamma^t \nabla \log \pi_\theta(a_t|s_t) Q_\theta^{T-t}(s_t, a_t)]\Big]$$

Actor-Critic Algorithms

- A Monte Carlo estimate of $\nabla V_\theta^T(s_0)$ is

$$g = \frac{1}{N}\sum_{i=1}^{N}\left(\sum_{t=0}^{T-1}\gamma^t \nabla \log \pi_\theta(a_{i,t}|\, s_{i,t})\hat{Q}_\theta^{T-t}(s_{i,t}, a_{i,t})\right)$$

where

$$\hat{Q}_\theta^{T-t}(s_{i,t}, a_{i,t}) = \sum_{k=t}^{T-1}\gamma^k R(s_{i,k}, a_{i,k}).$$

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where

$$\hat{Q}_\theta^{T-t}(s_{i,t}, a_{i,t}) = \sum_{k=t}^{T-1}\gamma^k R(s_{i,k}, a_{i,k}).$$

- Evidently, the Monte Carlo estimate is improved if the true value Q_θ^{T-t} instead of $\hat{Q}_\theta^{T-t}$ is used.

Introducing a "Critic"

- To estimate the value function $V_\theta^{T-t}(s)$, a regression approach with training data $\{y_{\ell,t}\}_{\ell=1}^N$ where

$$y_{\ell,t} \triangleq \sum_{k=t}^{T-1} \gamma^{k-t} R(s_{\ell,t}, a_{\ell,t})$$

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- A parametrized model $\hat{Q}_\phi^{T-t}(s,a)$ is obtained by solving

$$\min_\phi \sum_{\ell=1}^N \left\| y_{\ell,t} - Q_\phi^{T-t}(s_{\ell,t}, a_{\ell,t}) \right\|^2$$

Actor-Critic Algorithm

1. Given current parameter θ_n, generate samples

$$\tau_\ell = \{(s_{\ell,t}, a_{\ell,t})\}_{t=0}^{T-1}, \qquad \ell = 1, \ldots, N$$

   from policy π_{θ_n} and Markovian transition probabilities P.
2. Fit $\hat{V}_{\phi_n}^{T-t}$ to sampled reward sums, $t < T$
3. Compute

$$g(\theta_n) = \frac{1}{N} \sum_{\ell=1}^{N} \left(\sum_{t=0}^{T-1} \gamma^t \nabla \log \pi_{\theta_n}(a_{\ell,t} | s_{\ell,t}) \hat{Q}_{\phi_n}^{T-t}(s_{\ell,t}, a_{\ell,t}) \right)$$

   and $\theta_{n+1} = \theta_n + \alpha g(\theta_n)$
4. If $\|\theta_{n+1} - \theta_n\| < \epsilon$, Stop. Otherwise, back to 1.

Advantage Actor-Critic

- For reducing variance, we use $V_\theta^{T-t}(s_{\ell,t})$ as baseline so that

$$A_\theta^{T-t}(s_{\ell,t}, a_{\ell,t}) \quad = Q_\theta^{T-t}(s_{\ell,t}, a_{\ell,t}) - V_\theta^{T-t}(s_{\ell,t})$$

  measures relative performance of action $a_{\ell,t}$ when in state $s_{\ell,t}$ under continuation actions according to π_θ

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- The Monte Carlo estimate is

$$g(\theta_n) = \frac{1}{N} \sum_{\ell=1}^{N} \left(\sum_{t=0}^{T-1} \gamma^t \nabla \log \pi_\theta(a_{\ell,t} | s_{\ell,t}) A_{\phi_n}^{T-t}(s_{\ell,t}, a_{\ell,t}) \right)$$

Policy Gradient for $T = \infty$

For a randomized (parametrized) policy π_θ, the value function can be written as

$$V_\theta(s) = \sum_{a \in \mathcal{A}} \pi_\theta(a|s) Q_\theta(s,a) \tag{7}$$

where

$$Q_\theta(s,a) = R(s,a) + \gamma \sum_{s' \in \mathcal{S}} P(s'|s,a) V_\theta(s') \tag{8}$$

Policy Gradient (Infinite Horizon)

- Taking derivatives on (7) when evaluated in $s = s_0$ we obtain:

$$\nabla V_\theta(s_0) = \sum_{a \in \mathcal{A}} [\nabla \pi_\theta(a|s_0) Q_\theta(s_0, a) + \pi_\theta(a|s_0) \nabla Q_\theta(s_0, a)]$$

$$= \sum_{a \in \mathcal{A}} \left[\nabla \log \pi_\theta(a|s_0) Q_\theta(s_0, a) \pi_\theta(a|s_0) + \pi_\theta(a|s_0) \nabla Q_\theta(s_0, a)\right]$$

$$= \mathbb{E}_{a_0 \sim \pi_\theta(\cdot|s_0)} \left[\nabla \log \pi_\theta(a_0|s_0) Q_\theta(s_0, a_0)] + \nabla Q_\theta(s_0, a)\right] \tag{9}$$

and from (8)

$$\nabla Q_\theta(s_0, a) = \gamma \sum_{s_1 \in \mathcal{S}} P(s_1|s_0, a) \nabla V_\theta(s_1) \tag{10}$$

Policy Gradient (Infinite Horizon)

- Let

$$\phi(s) := \mathbb{E}_{a \sim \pi_\theta(\cdot|s)} \left[\nabla \log \pi_\theta(a|s) Q_\theta(s,a) \right]$$

Policy Gradient (Infinite Horizon)

- Let

$$\phi(s) := \mathbb{E}_{a \sim \pi_\theta(\cdot|s)} \left[\nabla \log \pi_\theta(a|s) Q_\theta(s,a) \right]$$

- Substituting (4) into (3) we obtain:

$$\nabla V_\theta(s_0) = \phi(s_0) + \gamma \sum_{a \in \mathcal{A}} \sum_{s_1 \in \mathcal{S}} \pi_\theta(a|s_0) P(s_1|s_0,a) \nabla V_\theta(s_1)$$

Policy Gradient (Infinite Horizon)

- Let
$$\phi(s) := \mathbb{E}_{a\sim\pi_\theta(\cdot|s)}\left[\nabla \log \pi_\theta(a|s) Q_\theta(s,a)\right]$$
- Substituting (4) into (3) we obtain:
$$\nabla V_\theta(s_0) = \phi(s_0) + \gamma \sum_{a\in\mathcal{A}} \sum_{s_1\in\mathcal{S}} \pi_\theta(a|s_0) P(s_1|s_0,a) \nabla V_\theta(s_1)$$
- In vector notation, $(I - \gamma \pi_\theta \circ P)\nabla V_\theta = \phi$ or equivalently:
$$\nabla V_\theta = (I - \gamma \pi_\theta \circ P)^{-1}\phi$$

Policy Gradient (Infinite Horizon)

- Recall that:

$$(I - \gamma \pi_\theta \circ P)^{-1} = I + \gamma(\pi_\theta \circ P) + \gamma^2(\pi_\theta \circ P)^2 + \gamma^3(\pi_\theta \circ P)^2 + \cdots$$

- The (s', s) entry (say $\rho_\theta(s', s)$ is the expected discounted total number of visits to s starting from s'. It follows that:

$$\begin{aligned}
\nabla V_\theta(s') &= \sum_{s \in \mathcal{S}} \rho(s', s)\phi(s) \\
&= \sum_{s \in \mathcal{S}} \mathbb{E}_{a \sim \pi_\theta(\cdot|s)}\left[\rho(s', s)\nabla \log \pi_\theta(a|s) Q_\theta(s, a)\right] \\
&= \sum_{s \in \mathcal{S}} \sum_{a \in \mathcal{A}} \rho(s', s)\nabla \log \pi_\theta(a|s) Q_\theta(s, a) \\
&= \mathbb{E}_{s \sim d_\theta(s')}\left[\nabla \log \pi_\theta(a|s) Q_\theta(s, a)\right]
\end{aligned}$$

where $d_\theta(s')$ is the normalized discounted visitation measure.